

Ahmed Nabil Reda Abdelkhalek

GIS Expert | GIS Project Manager | Professional Geospatial Services

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Portfolio: ahmednabilreda-igtm.github.io

PROFESSIONAL SUMMARY

Results driven GIS and Geospatial Systems Expert with over 20 years of proven experience in enterprise GIS implementation, spatial data infrastructure (SDI), and digital transformation within the urban development and housing sectors. Skilled in geospatial project management, data governance, and enterprise database administration. Successfully led national-scale programs supporting smart city development, digital twin initiatives, and land management systems across Saudi Arabia.

CAREER OBJECTIVE

To utilize my expertise in GIS project management, enterprise geospatial systems integration, and data governance frameworks to drive digital transformation, improve spatial data quality, and enhance decision-making processes for forward-thinking organizations.

PROFESSIONAL EXPERIENCE

GIS Expert | Geospatial Systems Integration

2023 – Present

Ministry of Municipalities and Housing (MOMAH) | Riyadh, Saudi Arabia

- Led enterprise geospatial data management and GIS operations across the Urban Planning & Land Agency, covering data governance, quality, database administration, and geospatial service delivery.
- Provided technical leadership for enterprise GIS initiatives, translating business requirements into scalable geospatial architectures, data models, workflows, applications, and integrated digital services.
- Directed the design, administration, and optimization of enterprise spatial databases and geospatial data models supporting Urban Maps Platform (UMAPS) & Enterprise Land and Asset Management System (ELAMS), with emphasis on performance, security, data integrity, access control, interoperability, and governance.
- contributed to developing geospatial strategies and implementation roadmaps aligned with digital transformation objectives, enterprise business requirements, and national geospatial and digital transformation frameworks.
- Led multidisciplinary geospatial teams and coordinated data collection, migration, integration, quality improvement, database operations, and enterprise GIS services across multiple business functions and government entities.
- Established and optimized geospatial data governance and quality processes, strengthening data lifecycle management, standardization, validation, integration, and enterprise data reliability.

- Provided technical advisory services for municipal asset geospatial data quality programs, supporting data assessment, standardization, governance, integration, and continuous quality improvement.
- Led geospatial requirements and spatial use cases for an AI-powered Urban Planning Agent, covering urban diagnostics, land-use change, idle land detection, urban growth forecasting, zoning recommendations, spatial optimization, policy simulation, and planning decision support.
- Provided geospatial technical leadership for the Digital Critical Sites Map for Flood and Rainfall Risk Mitigation, supporting the integration, analysis, and visualization of critical locations in coordination with emergency and government stakeholders.
- Directed geospatial data migration and integration across enterprise platforms, databases, and applications, ensuring data integrity, interoperability, business continuity, and efficient transition between systems.
- Supported urban planning and land-management applications through enterprise GIS, spatial databases, geospatial analytics, and location intelligence, enabling data driven planning and development decisions.

GIS Manager | Enterprise Geospatial Data Manager

2019 – 2023

National Housing Services Co. (NHSC) | Riyadh, Saudi Arabia

- Led enterprise GIS and geospatial data management projects, overseeing database development, national-scale geospatial data collection, migration, integration, and standardization in alignment with unified spatial data requirements.
- Managed GIS digital transformation and strategic planning initiatives, aligning enterprise GIS capabilities, geospatial services, technology roadmaps, and stakeholder requirements with organizational business objectives.
- Led the development and implementation of geospatial data governance frameworks and standards, covering data governance, geospatial coding systems, metadata, spatial data standards, data management processes, and data quality requirements.
- Provided technical leadership for enterprise geodatabase projects, supervising teams responsible for business requirements analysis, geospatial data modeling, database design, technical implementation, performance monitoring, and operational support.
- Directed the development and review of strategic geospatial reports and frameworks, including geospatial data collection strategies, data governance frameworks, geospatial coding systems, metadata standards, and enterprise spatial data standards.
- Managed stakeholder engagement and business requirements analysis, translating business needs into technical GIS requirements, data models, workflows, and enterprise geospatial solutions.
- Established quality assurance and technical governance controls to ensure project deliverables complied with approved technical specifications, business requirements, geospatial standards, and defined quality benchmarks.
- Performed geospatial data analysis, modeling, and optimization, supporting enterprise database development, data integration, standardization, and improved utilization of geospatial information.

GIS Consultant | Spatial Analysis & Land Information Systems

2015 – 2018

Saudi Arabia Parsons Ltd (SAPL), Ministry of Housing | Riyadh, Saudi Arabia

- Led GIS data collection and processing for the inventory of housing lands under the Idle Land Tax Initiative.

- Developed evaluation spatial models to support idle land spatial assessment and land related decision making.
- Performed spatial analysis of GIS data to support project requirements and urban land assessment.
- Managed geospatial data integrity, validity, and quality assurance throughout project workflows.
- Supervised GIS teams, delegated technical tasks, and supported the development of team capabilities.
- Identified, promoted, and advised on GIS solutions aligned with project requirements and business needs.
- Ensured GIS project deliverables met approved quality standards, technical specifications, and project requirements.

GIS Analyst | Urban Observatory & Municipal GIS

2011 – 2015

Zuhair Fayez Partnership (ZFP) | Makkah & Riyadh, Saudi Arabia

- Performed geospatial data analysis and processing to support Makkah Urban Observatory Project and Development of Geographic Information Systems for municipalities along the western axis of the Riyadh region.
- Conducted analytical data quality reviews, data extraction, loading, transformation, organization, and cleanup.
- Contributed to the development of reliable and robust GIS solutions as part of multidisciplinary project teams.
- Prepared cartographic mapping documents for publication and project deliverables.
- Produced detailed and accurate geospatial data reports to support project documentation and decision making.

GIS Team Leader | Spatial Data & Regional Planning

2009 – 2011

Azmi Abdulhadi & Abdulla Al Moaibed Consulting Engineering Company (AMCDE) | Dammam, Saudi Arabia

- Led GIS team activities supporting the development of regional base map datasets for the Tabuk region.
- Supervised and delegated GIS tasks, coordinated team activities, and strengthened technical team capabilities.
- Performed spatial analysis and data processing to support planning and requirements.
- Ensured the integrity, validity, and quality of geospatial data.
- Coordinated with GIS teams to develop and implement GIS standards.
- Identified and advised on GIS and integrated solutions aligned with project requirements.
- Ensured project deliverables complied with approved quality standards and technical requirements.

GIS Specialist | Municipal Base Mapping & Data Quality

2007 – 2009

Atheeb Integrated Solutions Company (AISC) | Jeddah, Saudi Arabia

- Supported the maintenance and updating of municipal base map databases using IKONOS satellite imagery from 2006 and 2007.
- Managed teams responsible for converting CAD drawings into GIS ready maps, ensuring correct spatial positioning and standardized attribute information.
- Performed quality control on digitized and entered for the Jeddah Water Network Project.
- Established and applied quality assurance and quality control procedures for GIS data production.
- Supported Jeddah City Planning activities, including the conversion and preparation of planning drawings for GIS integration.

- Performed raster image rectification to ensure correct geospatial positioning.

GIS Technician | Geospatial Data Conversion & Quality Control

2005 – 2007

Global GeoBits – Member of Osman Group Co | Cairo, Egypt

- Performed geospatial data conversion from hardcopy maps into digital GIS datasets for National Digital Land Information System (NDLIS) project.
- Supported the georeferencing of maps and preparation of spatial data for GIS workflows.
- Coordinated and distributed technical tasks among GIS production team members.
- Conducted quality control and data assembly activities to ensure the accuracy and consistency of GIS deliverables.
- Supported the preparation and organization of geospatial datasets for subsequent analysis and mapping.

EDUCATION

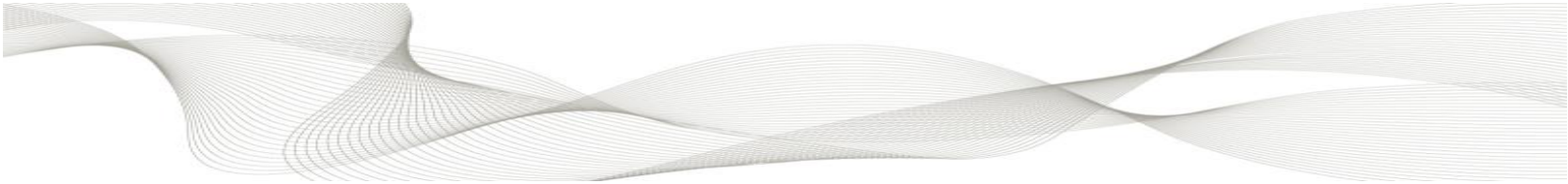
- Master of Science in GIS – Salzburg University, Austria | 2016 – 2019 | Graduated with Distinction
- High Diploma in GIS – Ain Shams University, Egypt | 2003 – 2005 | Very Good
- Bachelor of Arts in Geography – Ain Shams University, Egypt | 1999 – 2003 | Good

CERTIFICATIONS

- Project Management Professional (PMP®), Project Management Institute – Valid until Sep 2029 (License Number 2800652)
- Claude Academy: AI Fluency for creative work, Anthropic – Oct 2026
- Google AI Professional, Google – Sep 2026
- Business Analysis Professional (CBAP), Khatib & Alami Company – 2023
- Leading Business Transformation, inology company – 2022
- Spatial Data Science, Esri Academy – 2022
- Learning ArcGIS Python Scripting, Esri Academy – 2020

TECHNICAL SKILLS

- GIS Platforms & Tools | ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online, ArcGIS Workflow Manager, ArcGIS Model Builder, Esri ArcGIS Solutions, GeoMedia, Global Mapper, AutoCAD Map.
- Geospatial Data Management & Integration | Enterprise Geodatabase Design, Spatial Data Infrastructure (SDI), Data Governance Frameworks, Data Quality Management, Metadata Standards, Data Reviewer, Data Migration, Spatial Modeling.
- Databases & Programming | Microsoft SQL Server, Oracle Spatial, PostgreSQL/PostGIS, Python for ArcGIS (ArcPy), SQL Scripting, ETL Processes, API Integration.
- Remote Sensing & Analysis | ERDAS Imagine, Raster and Vector Analysis, Satellite Imagery Interpretation, Digital Twin Integration, Raster Data Processing, Site Selection Modeling, Image Classification.
- Spatial Analysis & Visualization | Network Analysis, 3D Geo-Design, Spatial Analytics, Map Projections, Geospatial Dashboards, Web GIS, Story Maps.

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- Cloud & Emerging Technologies | Cloud GIS Deployment, Smart City Applications, Geospatial AI & ML, IoT Integration, Mobile GIS Solutions.

PROFESSIONAL SKILLS

- Leadership and Team Management
- Strategic Planning and Decision Making
- Effective Communication and Stakeholder Engagement
- Problem Solving and Analytical Thinking
- Adaptability and Time Management

AWARDS & RECOGNITIONS

- Recognition for establishment of GIS Center – Jeddah Municipality
- Recognition for GIS Center Development – Ministry of Housing
- Recognition for contribution to the Idle Lands Project
- Recognition for development of Enterprise Land & Asset Management System (ELAMS)

KEYWORDS

GIS, Enterprise GIS, Geospatial Systems Integration, Spatial Data Infrastructure (SDI), Enterprise Geodatabase Design, Geospatial Data Governance, Data Governance Frameworks, Data Migration & Integration, Metadata Standards, OGC/ISO Compliance, Urban Planning, Land Management, Land-Use Change Detection, Idle Land Identification, Urban Growth Forecasting, Zoning & Spatial Optimization, Digital Twin, Smart Cities, AI-Driven Planning & Decision Support, Geospatial AI (GeoAI), Location Intelligence, Remote Sensing, Satellite Imagery Interpretation, Raster & Vector Analysis, Spatial Analytics, Web GIS, Geospatial Dashboards, ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online, Esri Solutions, PostgreSQL/PostGIS, Oracle Spatial, Python (ArcPy), ETL Processes, Digital Transformation, Data Quality Management, Enterprise Architecture, Stakeholder Engagement, Project Management, PMP, Digital Transformation.

REFERENCES & CERTIFICATES AVAILABLE UPON REQUEST